

SJF vs Priority Scheduling Project

SJF vs Priority Comparison

Input

Gantt Charts

Results

Analysis & Conclusion

Scheduling Mode

☐ SJF Preemptive (SRTF)

☐ Priority Preemptive

Process Input Table

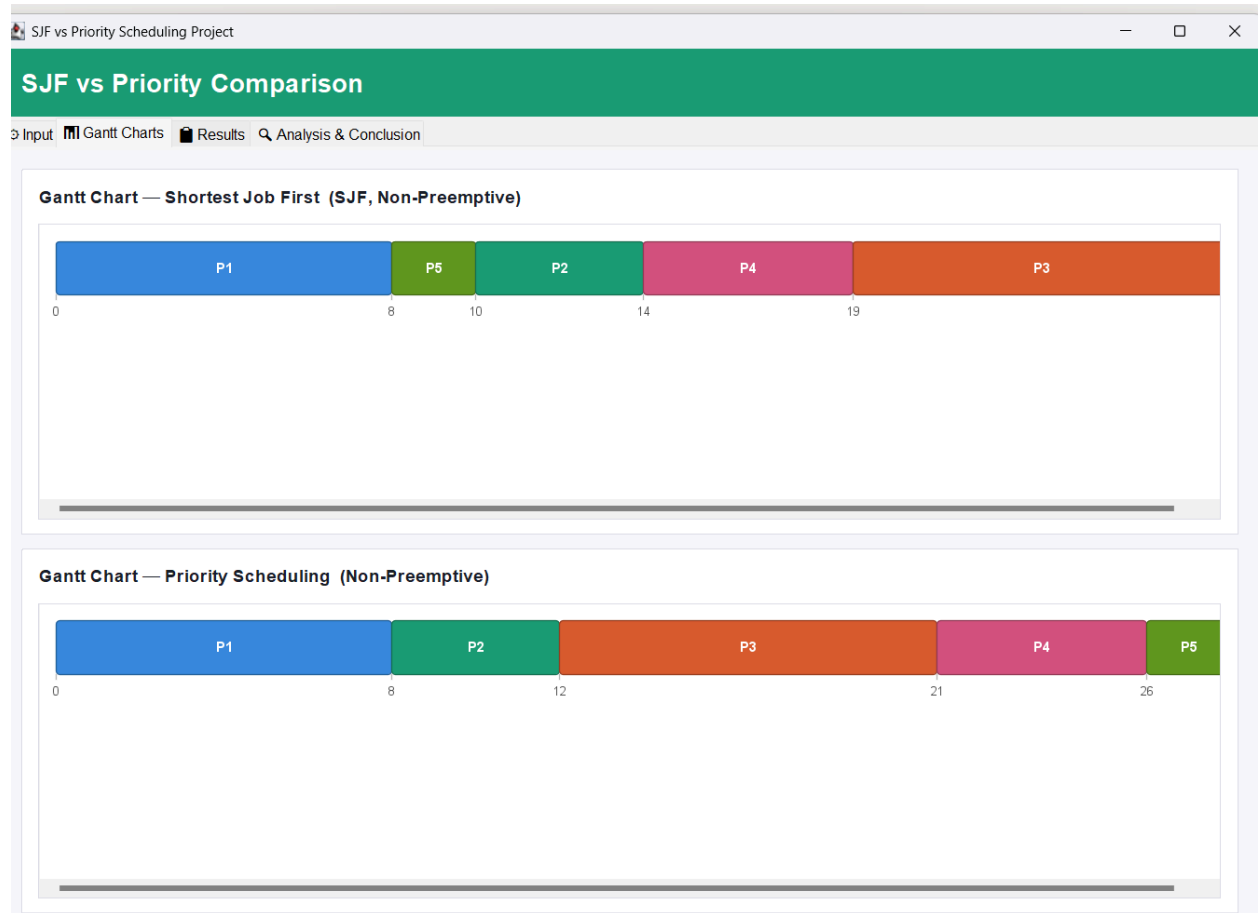
PID	Arrival Time	Burst Time	Priority
P1	0	8	2
P2	1	4	1
P3	2	9	3
P4	3	5	4
P5	4	2	5

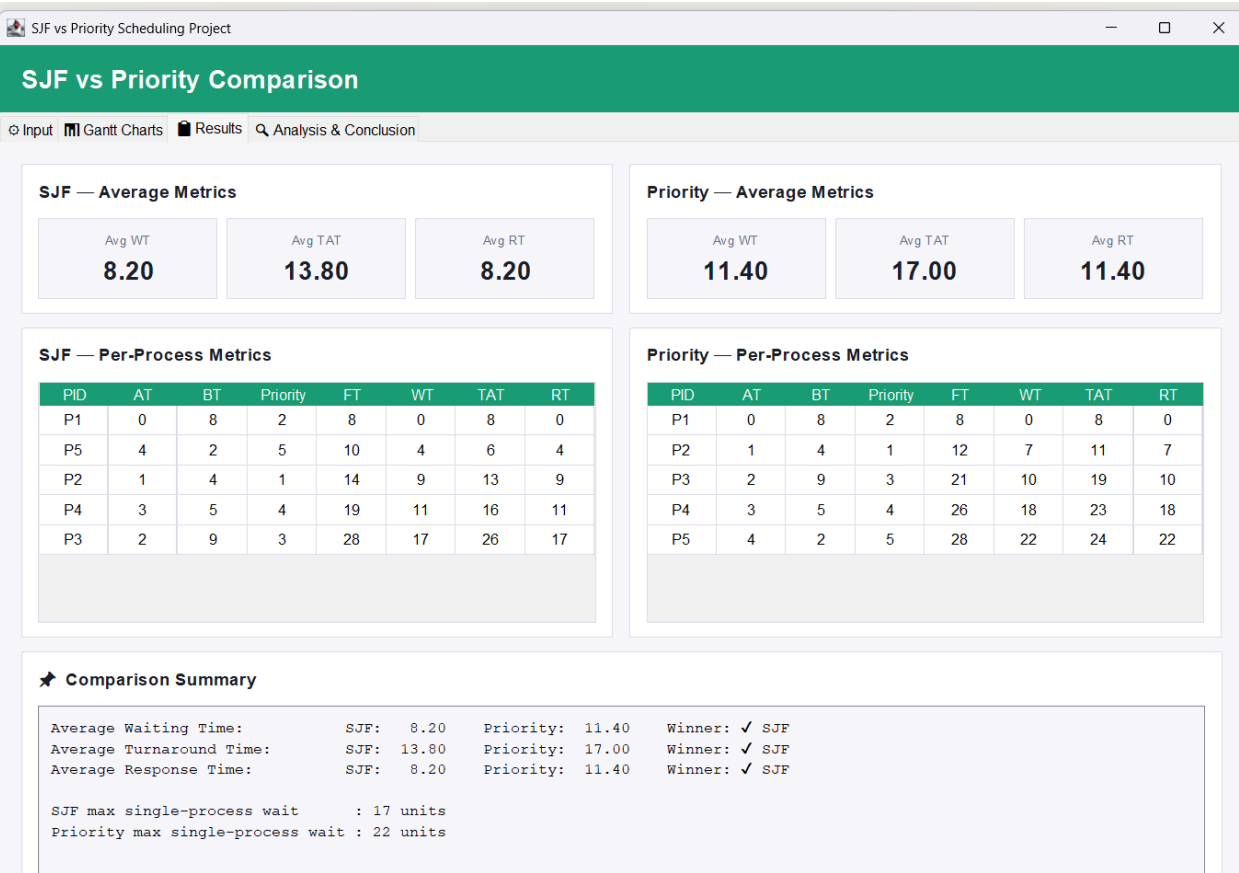
+ Add Process

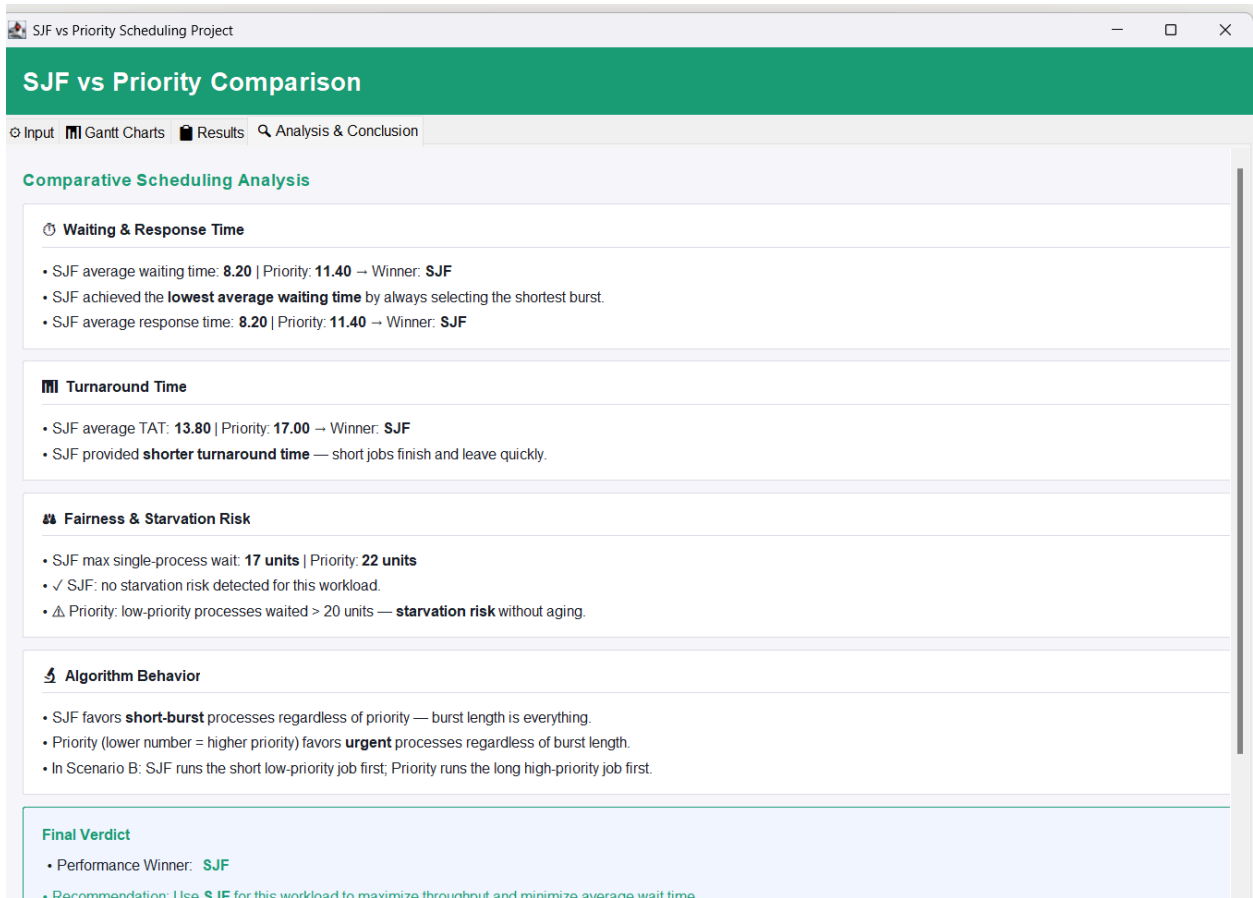
— Remove Selected

Clear All

▶ Run Simulation







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Process Input Table

PID	Arrival Time	Burst Time	Priority
P1	0	12	1
P2	2	2	5
P3	3	4	2
P4	5	6	3

+ Add Process

— Remove Selected

Clear All

▶ Run Simulation

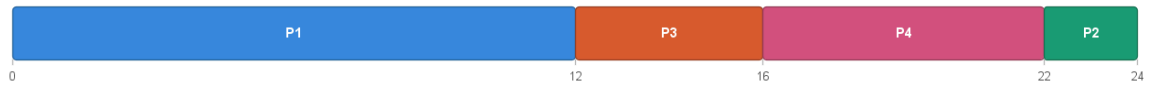
SJF vs Priority Comparison

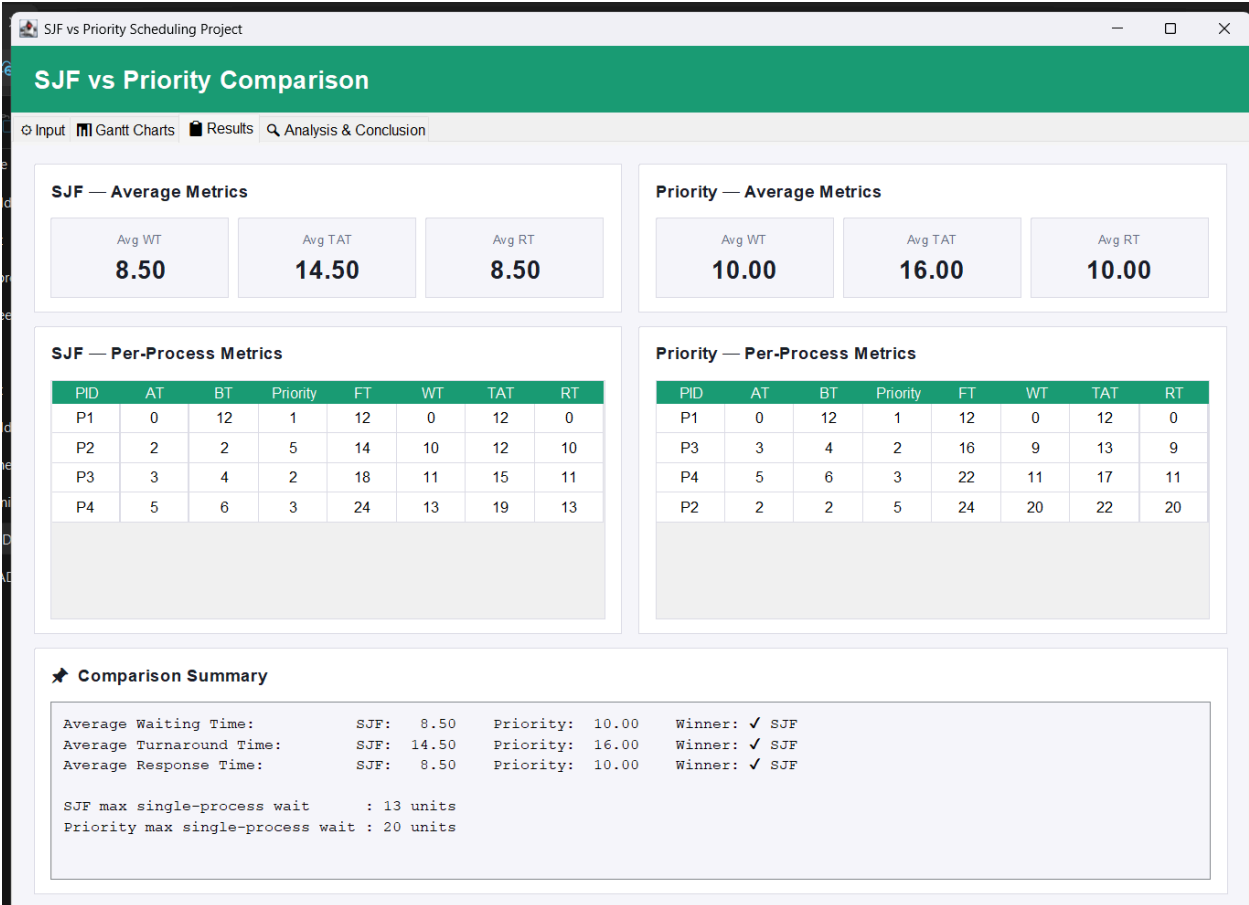
Input Gantt Charts Results Analysis & Conclusion

Gantt Chart — Shortest Job First (SJF, Non-Preemptive)



Gantt Chart — Priority Scheduling (Non-Preemptive)





SJF vs Priority Comparison

Input Gantt Charts Results Analysis & Conclusion

Comparative Scheduling Analysis

🕒 Waiting & Response Time

- SJF average waiting time: **8.50** | Priority: **10.00** → Winner: **SJF**
- SJF achieved the **lowest average waiting time** by always selecting the shortest burst.
- SJF average response time: **8.50** | Priority: **10.00** → Winner: **SJF**

📊 Turnaround Time

- SJF average TAT: **14.50** | Priority: **16.00** → Winner: **SJF**
- SJF provided **shorter turnaround time** — short jobs finish and leave quickly.

⚖️ Fairness & Starvation Risk

- SJF max single-process wait: **13 units** | Priority: **20 units**
- ✓ SJF: no starvation risk detected for this workload.
- ✓ Priority: no starvation risk detected for this workload.

🔗 Algorithm Behavior

- SJF favors **short-burst** processes regardless of priority — burst length is everything.
- Priority (lower number = higher priority) favors **urgent** processes regardless of burst length.
- In Scenario B: SJF runs the short low-priority job first; Priority runs the long high-priority job first.

Final Verdict

- Performance Winner: **SJF**
- Recommendation: Use **SJF** for this workload to maximize throughput and minimize average wait time.

SJF vs Priority Comparison

Input Gantt Charts Results Analysis & Conclusion

Scheduling Mode

☐ SJF Preemptive (SRTF) ☐ Priority Preemptive

Process Input Table

PID	Arrival Time	Burst Time	Priority
P1	0	5	1
P2	0	5	2
P3	0	12	5
P4	0	2	1
P5	0	3	2

+ Add Process

— Remove Selected

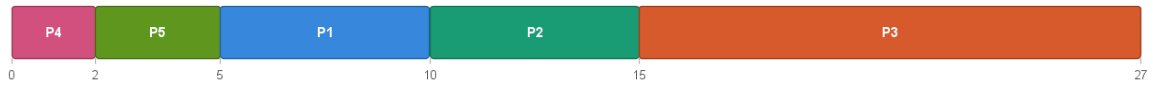
Clear All

▶ Run Simulation

SJF vs Priority Comparison

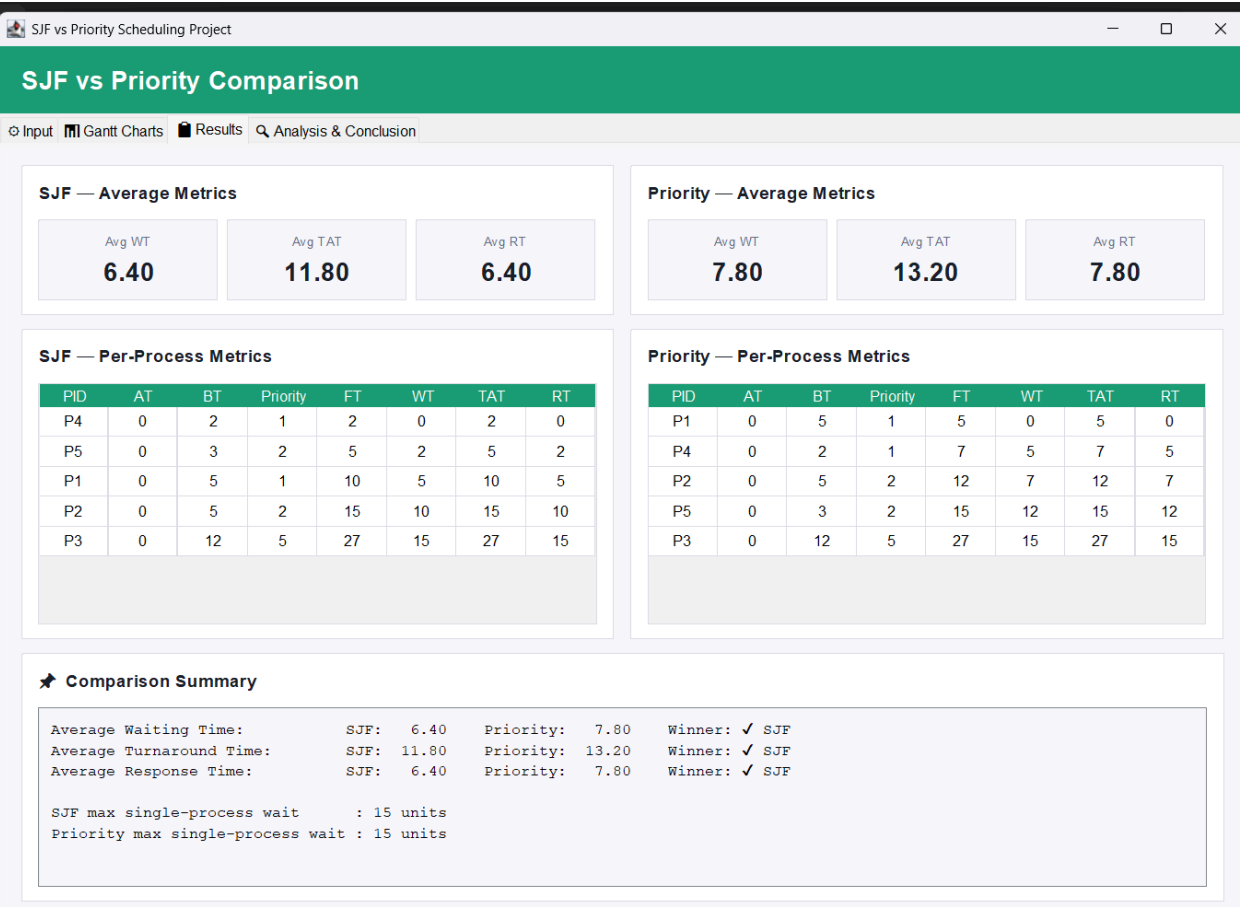
Input Gantt Charts Results Analysis & Conclusion

Gantt Chart — Shortest Job First (SJF, Non-Preemptive)



Gantt Chart — Priority Scheduling (Non-Preemptive)





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SJF vs Priority Comparison

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🕒

Waiting & Response Time

- SJF average waiting time: **6.40** | Priority: **7.80** → Winner: **SJF**
- SJF achieved the **lowest average waiting time** by always selecting the shortest burst.
- SJF average response time: **6.40** | Priority: **7.80** → Winner: **SJF**

📊

Turnaround Time

- SJF average TAT: **11.80** | Priority: **13.20** → Winner: **SJF**
- SJF provided **shorter turnaround time** — short jobs finish and leave quickly.

⚖️

Fairness & Starvation Risk

- SJF max single-process wait: **15 units** | Priority: **15 units**
- ✓ SJF: no starvation risk detected for this workload.
- ✓ Priority: no starvation risk detected for this workload.

🔍

Algorithm Behavior

- SJF favors **short-burst** processes regardless of priority — burst length is everything.
- Priority (lower number = higher priority) favors **urgent** processes regardless of burst length.
- In Scenario B: SJF runs the short low-priority job first; Priority runs the long high-priority job first.

Final Verdict

- Performance Winner: **SJF**
- Recommendation: Use **SJF** for this workload to maximize throughput and minimize average wait time.

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Process Input Table

⚠ Process P3: Arrival Time must be ≥ 0 .

PID	Arrival Time	Burst Time	Priority
P1	0	2	1
P2	0	3	2
P3	-2	12	5
P4	0	2	1

+ Add Process

— Remove Selected

Clear All

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Process Input Table

⚠ Process P2: Burst Time must be an integer (got "abc").

PID	Arrival Time	Burst Time	Priority
P1	0	2	1
P2	0	abc	2
P3	-2	12	5
P4	0	2	1

+ Add Process

— Remove Selected

Clear All

▶ Run Simulation